FINAL PROJECT

Programming for Data Science

Tennis Analysis - pdp_group-10

GROUP MEMBERS

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WEB SCRAPING

- Web site(s) selected : **Tennis Abstract**
- Resources chosen :
  - **Player profiles**: name, age, current rank
  - **Match history**: date, tournament, surface, round, winner, loser, score, and ranking fields
  - **Performance** statistics: selected rows for Overall last 52 weeks, Hard, Clay, and Grass

WEB SCRAPING

01

Shared utility module

- Set up a **headless Selenium Chrome driver**
- Normalize player names and text fields
- Retrieve the current **ATP Top-20 player list**

02

Dedicated scraping scripts

- Scrape player profiles
- Scrape match histories
- Scrape surface-specific performance statistics

03

Automated page loading and parsing

- Requests + lxml for static content
- Selenium for dynamic pages requiring interaction

04

Extracted data

- Stored in CSV files, one per resource
- Used as input for data cleaning and analysis

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WEB SCRAPING

Data from web scraping

- **top20_players.csv**
- matches_raw.csv
- stats_raw.csv

Data preparation

- Match records are reshaped into a consistent, analysis-ready format
- Statistics columns are renamed to clear, readable names and parsed into numeric values

→ produced: **matches_clean.csv, stats_clean.csv**

PROJECT GOALS

1. How important is ranking for match outcomes?
2. Do players perform differently depending on the surface?
3. Which player characteristics are associated with winning matches?

INITIAL EVALUATION

A data quality inspection is performed:

- **Missing data** found:
  - in matches, one match where the loser was unranked at match time
  - in stats, players who never played on a given surface (e.g. grass)
  - tiebreak statistics, players played on a surface but no tiebreaks occurred
- **Duplicate matches**: same match was scraped twice, once per player perspective
- Percentage columns read as strings due to % suffix (e.g. "79.8%")

DATA CLEANING

- Removed duplicate matches (kept one occurrence)
- Stripped % suffix from percentage columns and cast to float
- Missing values were preserved since they reflect lack of observation rather than poor performance.

Data validation was also perfomed to:

- verify no duplicate match rows remain
- confirm stats percentage fields are floats

Q1. HOW IMPORTANT IS RANKING FOR MATCH OUTCOMES?

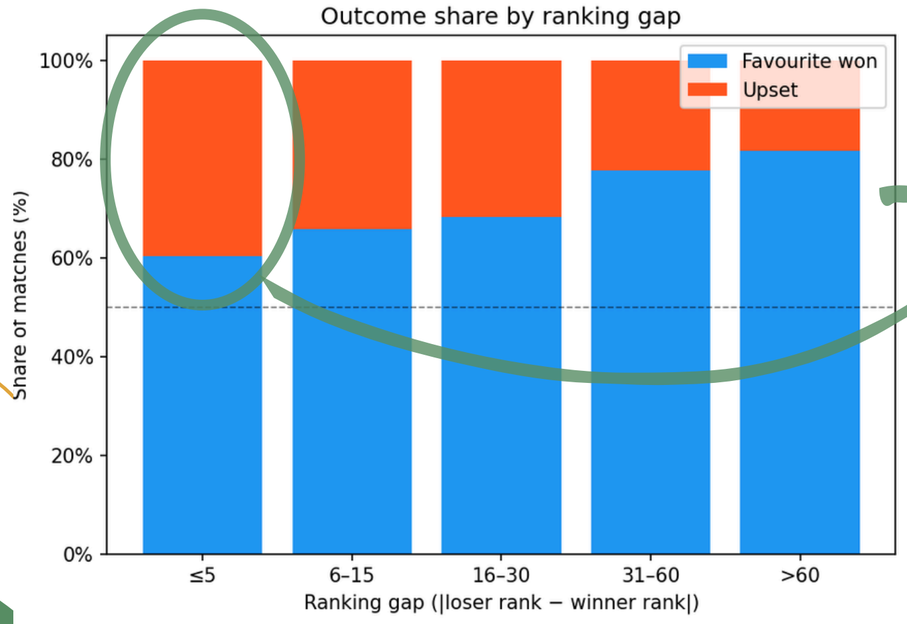
Data tasks

- **Excluded** one match with **missing ranking**
- Computed **ranking difference** between winner and loser
- **Classified matches** as "favourite won" or "upset"
- **Grouped matches** by ranking gap ranges

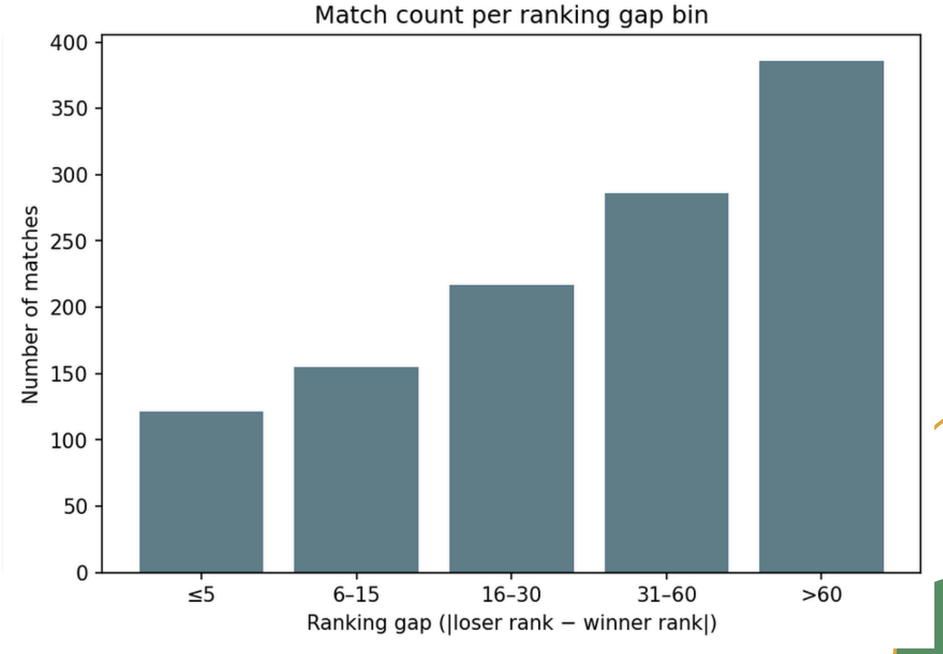
```
Matches analysed: 1165
Favourite won: 860 (73.8%)
Upsets: 305 (26.2%)
```

The higher-ranked player wins most matches (~74%)

Q1. HOW IMPORTANT IS RANKING FOR MATCH OUTCOMES?



Upsets are common when players are similarly ranked



Large ranking gaps strongly reduce upset probability

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Q2. DO PLAYERS PERFORM DIFFERENTLY DEPENDING ON THE SURFACE?

Data tasks

- **Filtered stats** to Hard, Clay and Grass surfaces
- Computed matches played per player and surface (matches = wins + losses), used to flag small-sample surface results
- **Preserved missing values** where players did not play on a surface
- Computed **per-player surface consistency**

Q2. DO PLAYERS PERFORM DIFFERENTLY DEPENDING ON THE SURFACE?

Win rate by player and surface (ATP Top-20, 2025-26)
(n = total matches played)

Carlos Alcaraz	89% n=48	95% n=21	92% n=12
Jannik Sinner	91% n=63	87% n=15	89% n=9
Novak Djokovic	83% n=25	82% n=11	83% n=7
Alexander Bublik	67% n=43	83% n=31	83% n=6
Alexander Zverev	70% n=58	81% n=22	67% n=9
Casper Ruud	59%	75%	N/A

→ The most successful players perform well across surfaces, with very low cross-surface variability.

Surface dependence is stronger for mid-tier players, but is also more sensitive to match volume →

Tommy Paul	n=27	n=19	n=3
Frances Tiafoe	61% n=40	60% n=15	33% n=3
Lorenzo Musetti	63% n=45	76% n=18	0% n=1
Francisco Cerundolo	57% n=28	67% n=30	0% n=2
	Hard	Clay	Grass

Casper Ruud	59% n=36	75% n=21	N/A n=0
Alex De Minaur	70% n=20	69% n=4	60% n=2
Taylor Fritz	65% n=56	43% n=7	87% n=18
Ben Shelton	68% n=44	62% n=14	60% n=10
Jiri Lehecka	64% n=45	55% n=12	70% n=11
Karen Khachanov	53% n=33	60% n=16	73% n=12
Daniil Medvedev	73% n=57	50% n=9	63% n=8
Flavio Cobolli	54% n=39	65% n=18	67% n=9
Felix Auger Aliassime	74% n=17	50% n=2	60% n=2
Andrey Rublev	59% n=41	57% n=18	67% n=6
Alejandro Davidovich Fokina	60% n=9	50% n=5	71% n=2
Tommy Paul	65% n=27	83% n=19	33% n=3

Grass results are least reliable due to small samples.

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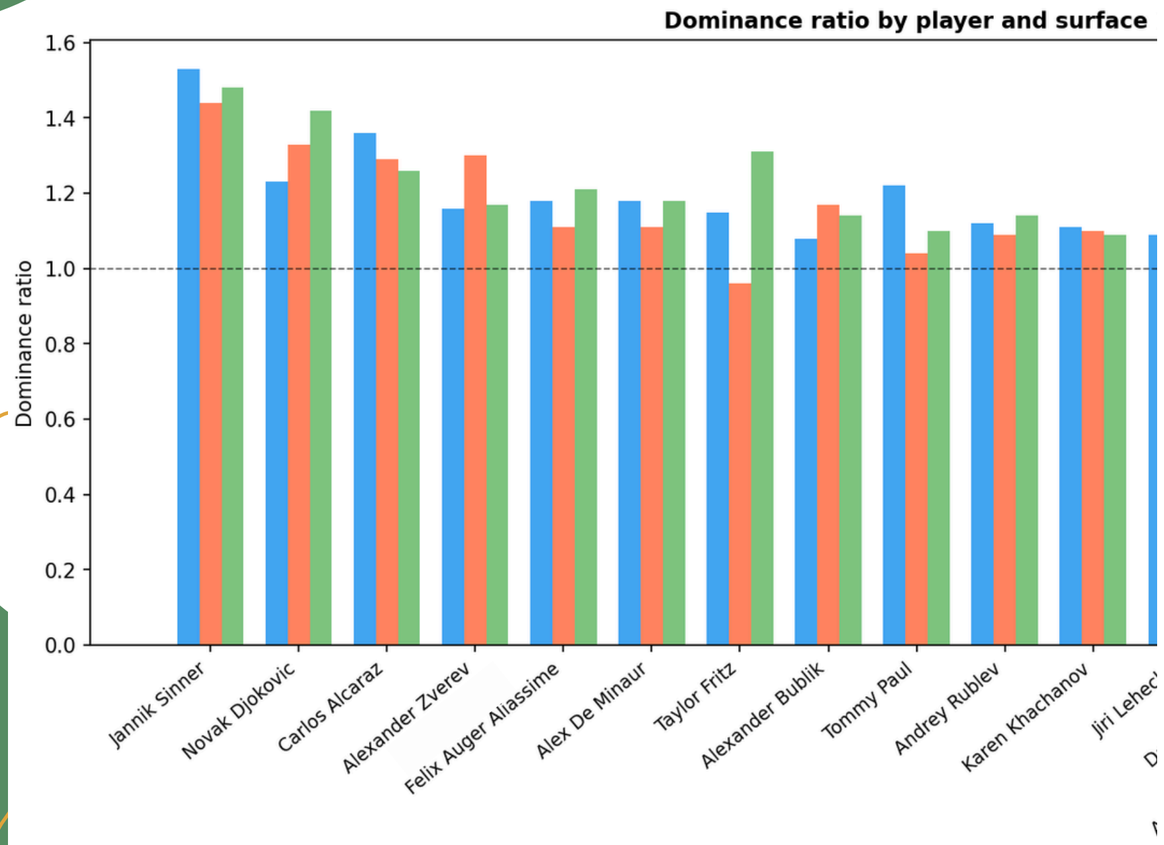
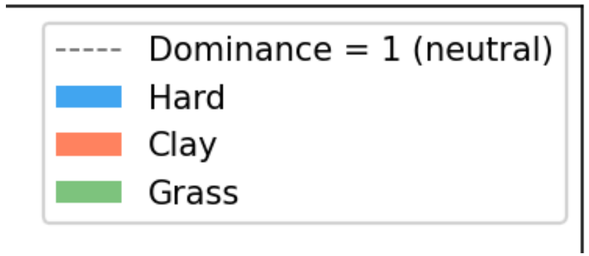
Q2. DO PLAYERS PERFORM DIFFERENTLY DEPENDING ON THE SURFACE?





- Dominance ratio confirms the strongest surface patterns seen in win rates
- Where both metrics peak on the same surface, the surface effect is structural

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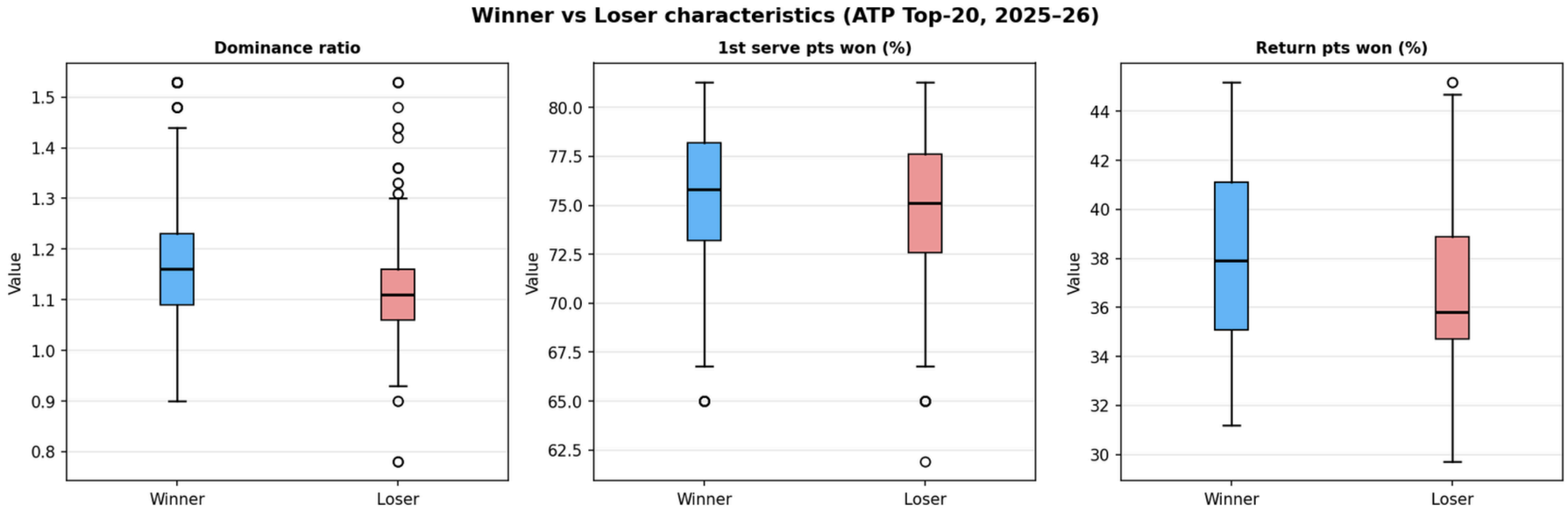
Q3. WHICH PLAYER CHARACTERISTICS ARE ASSOCIATED WITH WINNING?

Data tasks

- **Joined matches with** surface-specific **stats** for winners and losers
- **Joined player age and ranking** data
- Computed **winner–loser differences** for each metric
- **Preserved missing values** when stats were not observed

Q3. WHICH PLAYER CHARACTERISTICS ARE ASSOCIATED WITH WINNING?



- **Winners** show **higher dominance ratio**, indicating superior point control.
- **First serve points** won is **higher for winners**, not just serve frequency.
- **Return points** won clearly **separates winners** from losers.

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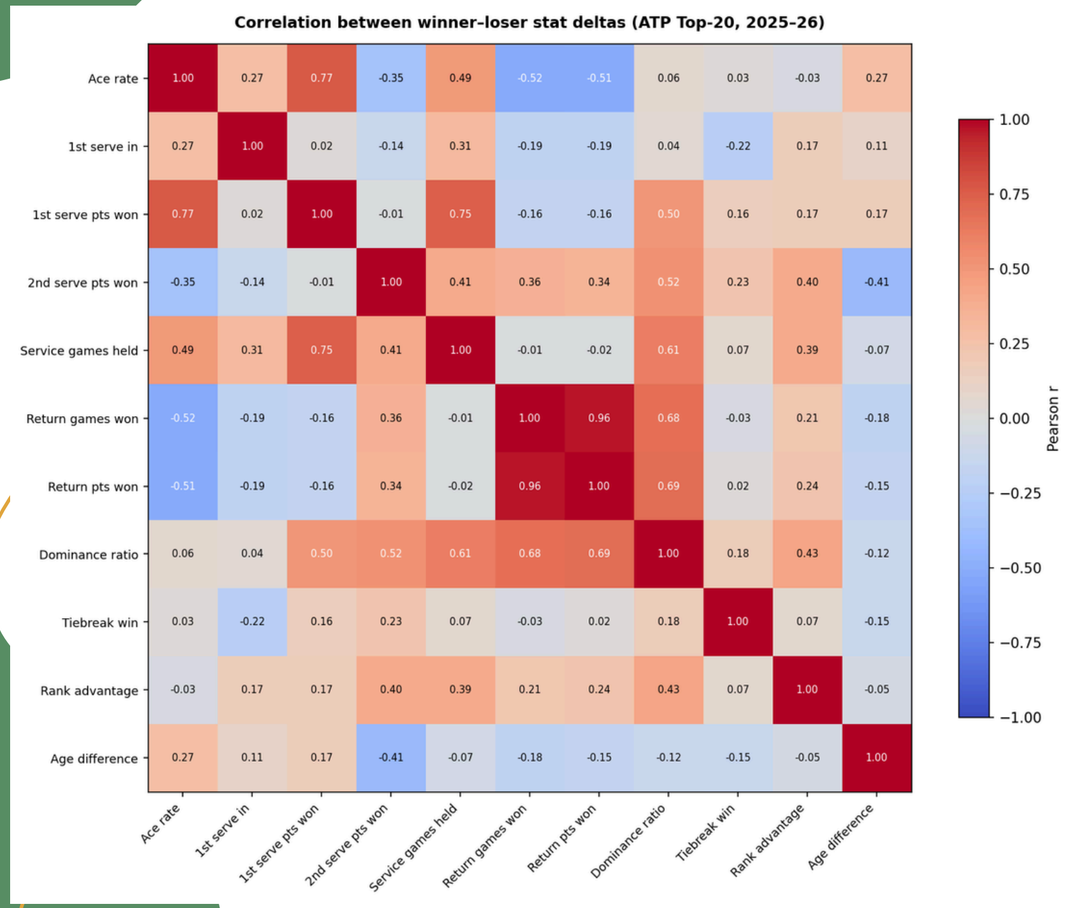
Q3. WHICH PLAYER CHARACTERISTICS ARE ASSOCIATED WITH WINNING?



- Return games won and return points won are almost perfectly correlated
- Dominance ratio strongly correlates with return performance
- First serve points won strongly correlates with service games held
- Ace rate correlates with serve metrics but weakly with match success

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CONCLUSION

- **Ranking matters**, but it is not absolute: higher-ranked players win most matches, yet upsets remain common when ranking gaps are small.
- **Surface effects are real**, especially below the very top level: elite players perform consistently across surfaces, while mid-tier players show stronger surface dependence.
- **Match outcomes are driven by point-level performance**, not single metrics: dominance ratio and return effectiveness best explain why players win.
- Overall, success at the top level reflects **balanced, all-court performance**, rather than specialization in isolated skills.

THANK YOU

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